

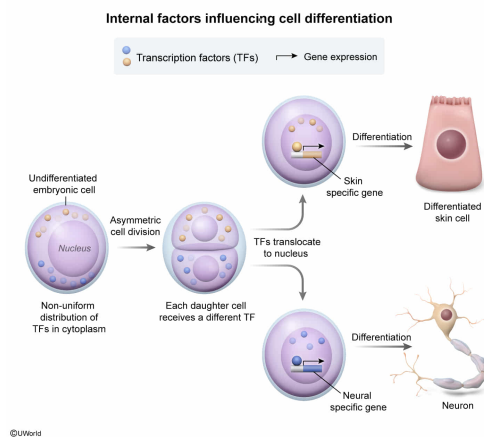
Cell differentiation during embryonic development is hypothesized to be controlled by differences in gene expression due to external and internal factors. Which experimental observation best supports a mechanism of cell differentiation that is due to internal factors?

- ☐ A. A gradient of signaling molecules triggers a cascade, leading to new gene expression in embryonic cells. [21%]
- ✓ ☒ B. Unequal distribution of transcription factors across embryonic cells leads to differential gene expression. [64%]
- ☐ C. Cell-to-cell contact induces new gene expression in an adjacent cell. [9%]
- ☐ D. Differences in oxygen levels repress expression of certain genes. [5%]

Correct

63% answered correctly

Explanation:



Following **fertilization** in multicellular organisms, the single-celled zygote is **totipotent**, or able to develop into any of the cells or tissues of the organism. Once cell division begins, the resulting **embryonic** cells become progressively restricted in their developmental fates, a process known as cell determination. For example, during **gastrulation** the three germ layers (**ectoderm**, **mesoderm**, and **endoderm**) are formed. Afterward, ectodermal cells are restricted to becoming either surface ectoderm, **neural crest**, or neural tube.

Once the cell's fate has been determined, a program of **cell differentiation** allows for the formation of specialized tissues and cell types. During cell differentiation, **differential gene expression** patterns result in the specific features (eg, size, morphology, metabolic characteristics) of that cell type. For example, ectodermal cells specialize into specific cell types within that germ layer (eg, epidermis, neurons, melanocytes).

The question asks which observation best supports a mechanism of cell differentiation due to internal factors in the cell. Differences in **gene expression**, due to both internal and external factors, drive the cell differentiation process. Internal factors do not depend on interactions of cells with each other or with their environment, and result from inherent differences among individual cells. External factors influencing cell differentiation involve interactions with neighboring cells and the environment surrounding the cell.

Asymmetric distribution of **transcription factors** to embryonic cells during **cell division** is a **mechanism of cell differentiation** due to an **internal factor**. Other examples of internal factors include **epigenetic modifications** such as **DNA methylation** or **chromatin** remodeling.

(Choices A, C, and D) These choices represent external (not internal) mechanisms of cell differentiation. **Direct contact** between adjacent cells and secretion of short- and longer-range **signaling molecules** trigger **signaling cascades** that result in differential expression or repression of certain **genes**. In addition, environmental effects (eg, differences in oxygen levels) can differentially affect gene expression as well.

Educational objective:

In multicellular organisms, many distinctive cell types must be formed from a single-celled zygote. During development, both external (eg, secreted signals, environmental cues) and internal factors (eg, chromatin remodeling, DNA methylation, asymmetric distribution of cellular components) influence gene expression in a differential manner, which culminates in the differentiation of cells into a variety of cell types.

Time spent:0 sec**Subject:**

Biology

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